

Smart Cat Litter *Box Monitor*



Passive health tracking for cats through visit, weight, and pattern monitoring.

TEAM

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TEAM

Team SmartLitterBox

GITHUB

[https://github.com/MengqiS21/
TECHIN515_final_project_smart_litter_box](https://github.com/MengqiS21/TECHIN515_final_project_smart_litter_box)

The Problem & Who It Affects

THE PROBLEM

Cat owners have no easy way to detect early signs of health issues from litter box behavior.

Cats often hide symptoms of illness such as urinary blockages or weight loss. Subtle changes in litter box visit frequency, duration, and body weight are hard to notice without consistent tracking. By the time owners spot a problem, the condition may already be serious.

BY THE NUMBERS

64.5%

of cats with urinary tract issues show no obvious symptoms until the condition becomes serious

INTENDED USER GROUP

Cat owners who want passive, automatic monitoring of their cat's health through litter box usage patterns, without relying on manual observation or frequent vet visits.

Our Solution & Key Features

IN ONE SENTENCE

A litter box mounted edge device that uses a PIR sensor and load cell to track each visit's duration and weight, runs anomaly detection against a personalized baseline on the ESP32S3, and syncs logs to a cloud dashboard for long term trend viewing.

HARDWARE STACK

XIAO ESP32S3 | Load cell + HX711 | PIR sensor

KEY FEATURES

01

Visit Detection

PIR sensor captures entry and exit automatically for every visit.

02

Weight Tracking

Load cell measures cat weight per visit and tracks long term trends.

03

On Device Anomaly

Rule based detector compares each visit to a personalized 7 day baseline.

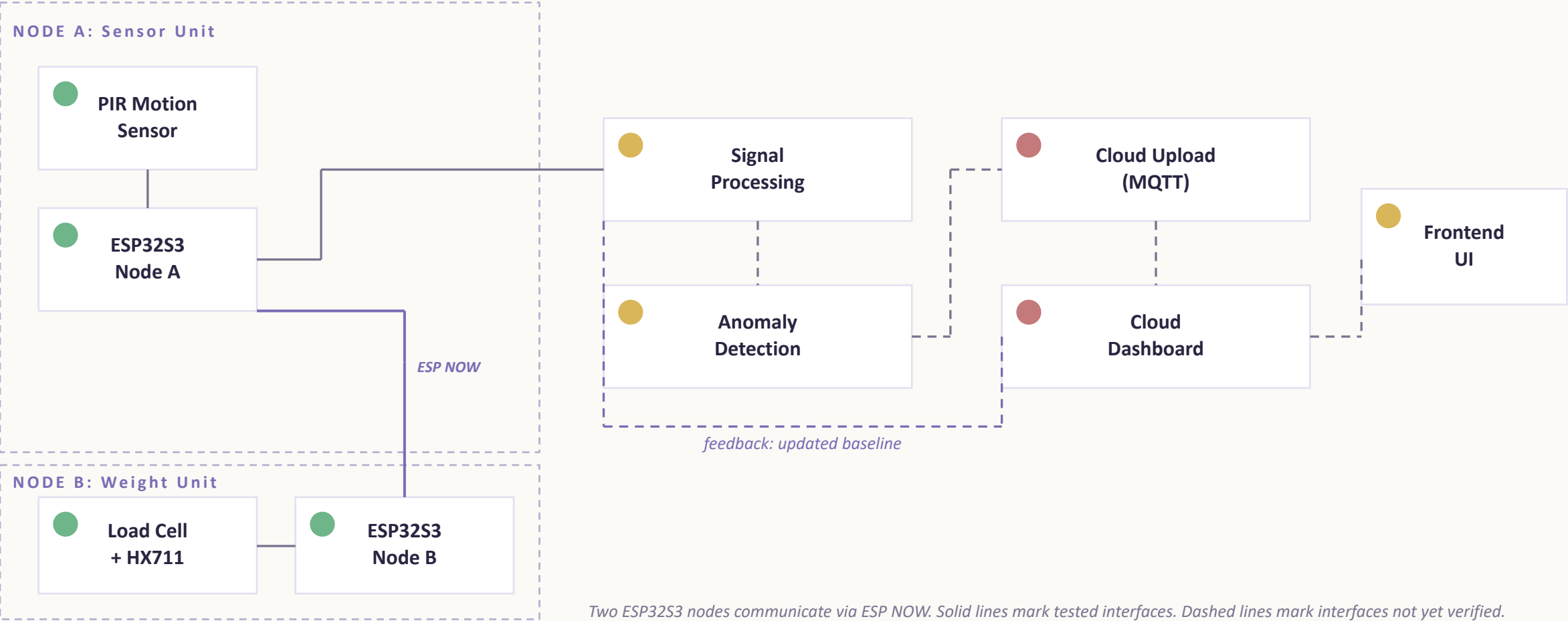
04

Cloud Trends

MQTT upload to a dashboard so owners can view history and alerts.

Updated System Architecture

Proven In progress Blocked



Two ESP32S3 nodes communicate via ESP NOW. Solid lines mark tested interfaces. Dashed lines mark interfaces not yet verified.

Trade Off Analysis

Sweeping system parameters to evaluate impact on performance metrics.

01

Stabilization Window

PARAMETER

HX711 stabilization time controls how long the system waits after detecting a visit before recording a stable weight reading. Tested at 1s, 3s, 5s, and 7s windows to evaluate the trade off between reading accuracy and response speed.

RESULT

At 1s, readings showed high variance due to the cat's movement on the load cell. At 3s, standard deviation dropped significantly but occasional spikes remained. At 5s, mean error reached 0.11 g with 0.23 g standard deviation, providing stable readings with acceptable latency. At 7s, accuracy improved only marginally while adding unnecessary delay.

02

Communication Protocol

PARAMETER

Evaluated three options for communication between the sensor node (ESP32S3 + PIR) and the weight node (ESP32S3 + HX711): ESP NOW, Wi Fi, and BLE. Compared on latency, power consumption, and ease of implementation.

RESULT

Wi Fi offers high throughput but requires a router and draws significantly more power, which is not ideal for a battery powered device. BLE is low power but introduces pairing complexity and higher latency for real time data transfer. ESP NOW enables direct peer to peer communication between two ESP32S3 boards with no router dependency, sub 10 ms latency, and minimal power overhead.

Validation: Data Collection & Integration

Systematic validation of sensor integration, data pipeline, and model readiness.



Data Collected	Quantitative Results	Model / Next Steps
600 images total across three categories (Cat A, Cat B, Empty). All captured via PIR triggered camera on the ESP32S3. Insert sample images from each category and a photo of the hardware setup.	HX711 weight sensor validated with mean error 0.11 g and std dev 0.23 g over 20 trials. Image classification accuracy 89.8% on test set.	lassification model can distinguish between Cat A, Cat B, and Empty. Next step is integrating the classifier output with weight and visit duration data to build the full anomaly detection pipeline. Cloud upload and dashboard display to follow.

Budget Update

TOTAL BUDGET

\$ 350

SPENT SO FAR

\$ 57.38

REMAINING

\$ 292.62

PURCHASES TO DATE

Item	Qty	Unit Cost	Total
[LiPo battery]	[1]	[\$ 5.83]	[\$ 5.83]
[LiPo charger module (TP4057, 5pcs)]	[1]	[\$ 7.56]	[\$ 7.56]
[Cat litter box(Smart Paws,enclosed)]	[1]	[\$ 43.99]	[\$ 43.99]
[XIAO ESP32S3]	[2]	[\$ 0.00]	[\$ 0.00]
[Load cell + HX711]	[1]	[\$ 0.00]	[\$ 0.00]
[PIR motion sensor]	[1]	[\$ 0.00]	[\$ 0.00]